

Can Transformer Efficiency Approach 100% Under Ideal Conditions?

Physics Unit 4 Research Investigation

Claim evaluated: *Transformer efficiency can approach 100% under ideal conditions.*

Word count: approximately 1,994 words, excluding headings, tables and references

Claim

Transformer efficiency can approach 100% under ideal conditions.

Research question

To what extent do the loss mechanisms and published performance data for liquid-immersed distribution transformers support the claim that transformer efficiency can approach 100% under ideal conditions?

The quantitative evidence is drawn from the DOE's regulated transformer test method and efficiency standards, supported by a published 100 kVA product specification [1-3].

Rationale

A transformer transfers electrical energy between circuits through electromagnetic induction. A changing primary current produces changing core flux, which induces a secondary emf by Faraday's law. The key term *efficiency* is the fraction of input electrical power transferred as useful output power. The claim's phrase *approach 100%* requires a distinction between a value that becomes arbitrarily close to 100% and a value that equals it. Similarly, *ideal conditions* could mean the lossless theoretical model or merely well-controlled conditions for an actual transformer. These interpretations lead to different conclusions, so this investigation evaluates both rather than treating the claim as a simple statement of fact.

The relevant relationship is

$$\eta = \frac{P_{\text{out}}}{P_{\text{in}}} \times 100 = \frac{P_{\text{out}}}{P_{\text{out}} + P_{\text{loss}}} \times 100.$$

For an ideal transformer,

$$\frac{V_p}{V_s} = \frac{N_p}{N_s}, \quad V_p I_p = V_s I_s,$$

where V is rms voltage, I is rms current and N is the number of turns. The ideal model therefore has $P_{\text{loss}} = 0$, whereas a real transformer has $P_{\text{loss}} > 0$ and hence $\eta < 100\%$. Efficiency tends towards 100% only as the ratio $P_{\text{loss}}/P_{\text{out}}$ tends towards zero.

Initial research identified two principal loss groups in real transformers. **No-load (core) loss** occurs whenever rated voltage is applied: magnetic domains repeatedly realign, causing hysteresis loss, and changing flux induces eddy currents. **Load loss** increases with winding current. Its dominant component is copper loss, $P_{\text{Cu}} = I^2 R$, with additional stray eddy-current losses from leakage flux. Lamination, higher-resistivity core materials, increased conductor cross-section and improved geometry can reduce these losses but do not remove them from a conventional operating transformer [1, 4].

Liquid-immersed distribution transformers were selected because the DOE provides comparable efficiencies across many ratings at a prescribed 50% load [1, 2]. This controls transformer class and test condition while allowing capacity to be investigated as the first variable. The second variable is design-standard period: comparing the 2016-2029 and 2029 minimum efficiencies at the same rating tests whether practical engineering reduces the loss fraction. The third variable is fractional load, because fixed core loss and current-dependent load loss predict a non-linear efficiency relationship. A 100 kVA liquid-filled unit was selected for this calculation because its specification separately reports no-load and full-load loss [3]. These three variables, capacity, design standard and fractional load, provide complementary tests of whether the efficiency gap can become very small under increasingly favourable conditions.

The DOE separately measures no-load and load loss, corrects them to defined temperatures, then calculates total loss and efficiency [1]. At fractional load x , the model is

$$P_{\text{loss}}(x) = P_0 + x^2 P_{\text{FL}},$$

where P_0 is no-load loss and P_{FL} full-load loss. It follows the regulated requirement to adjust load losses to the specified output loading level [1]. It predicts maximum efficiency when $P_0 = x^2 P_{\text{FL}}$: core loss dominates below this load and $I^2 R$ loss above it.

Therefore, the research question asks whether theoretical loss mechanisms and the three selected practical comparisons jointly justify the wording of the original claim.

Evidence and analysis

Dataset 1: Present minimum efficiency versus rated capacity

Table 1 reproduces selected values from the current US minimum standard for liquid-immersed transformers made from 2016 until 22 April 2029, measured at 50% load [2]. It is nationally regulated data with a defined method and a wide rating range, although it is a compliance minimum rather than an installed-fleet mean.

Rated capacity / kVA	Minimum efficiency / %	Energy lost per 1000 kWh delivered / kWh
10	98.70	13.17
25	98.95	10.61
50	99.11	8.98
100	99.25	7.56
250	99.39	6.14
500	99.49	5.13
833	99.55	4.52

Table 1. Selected values from eCFR 10 CFR 431.196(b)(2); final column calculated using $E_{\text{loss}} = 1000(1/\eta - 1)$, with η expressed as a decimal [2].

Trend and relationship. At the fixed 50% load condition, minimum efficiency increases monotonically from 98.70% at 10 kVA to 99.25% at 100 kVA and 99.55% at 833 kVA. The corresponding calculated loss falls from 13.17 to 7.56 and then 4.52 kWh per 1000 kWh delivered. Thus, increasing capacity by a factor of 83.3 raises the regulated minimum by 0.85 percentage points, but even the largest listed unit remains 0.45 percentage points below 100%.

Limitation. These values are legal minimum efficiencies, not an installed-transformer sample. Ageing, waveform, temperature and load can alter field efficiency. The dataset provides a lower-bound comparison at one condition, but cannot determine average efficiency or a capacity-efficiency relationship.

Scientific argument. The decreasing loss fraction is consistent with the loss model. At a common fractional load, useful output scales with rating, while core and winding losses need not increase in direct proportion because a larger transformer can use a greater conductor cross-section and core mass relative to its power rating. Lower winding

resistance reduces I^2R loss, and lower loss divided by a larger output produces a higher η [4]. Yet a conventional core still has hysteresis and eddy-current loss, while finite-resistance windings still dissipate power; the numerator of the efficiency equation cannot equal the denominator.

Link to the research question. Capacity therefore provides qualified support for the claim: practical liquid-immersed transformers can become very efficient as the loss fraction falls, but this variable provides no empirical basis for saying that a real transformer reaches 100%.

Dataset 2: Effect of tighter standards on the same technology

Table 2 compares the present standard with that scheduled from 23 April 2029 for non-submersible liquid-immersed transformers [2]. Holding class and 50% load constant enables a valid comparison.

Rated capacity / kVA	2016-2029 minimum / %	2029 minimum / %	Increase / percentage points	Reduction in loss per 1000 kWh delivered / kWh
10	98.70	98.77	0.07	0.72
50	99.11	99.15	0.04	0.41
100	99.25	99.29	0.04	0.41
250	99.39	99.51	0.12	1.22
500	99.49	99.59	0.10	1.01
833	99.55	99.64	0.09	0.91

Table 2. Efficiencies from eCFR 10 CFR 431.196(b)(2) and (b)(3); energy reductions calculated from the two efficiency values [2].

Trend and relationship. Every selected rating has a higher future threshold. The increase is 0.04 percentage points at 50 and 100 kVA, 0.12 points at 250 kVA and 0.09 points at 833 kVA. For example, the 500 kVA threshold rises from 99.49% to 99.59%, reducing maximum loss from 5.13 to 4.12 kWh per 1000 kWh delivered, approximately 20%.

Limitation. The comparison is internally consistent because capacity, transformer class and test load are controlled, but the future values are prospective legal thresholds rather

than field measurements. The data cannot show which design change caused each improvement, or whether every transformer made after 2029 will operate at its threshold. It instead establishes the technically feasible minimum required under the prescribed test conditions.

Scientific argument and link to the research question. Reducing core-loss and load-loss components through improved magnetic materials, core laminations, conductor dimensions or geometry lowers P_{loss} relative to output, increasing η [1, 4]. However, the highest selected future value, 99.64%, still loses 3.61 kWh per 1000 kWh delivered. This variable consequently supports the proposition that engineering can move conventional transformers closer to 100%, but it also shows that improved practical standards do not demonstrate equality with 100%.

Dataset 3: Modelled effect of load using published loss components

The load dependence hidden by the 50% test condition is evaluated using a 100 kVA, single-phase liquid-filled transformer. The supplier reports $P_0 = 273$ W and $P_{\text{FL}} = 1105$ W [3]. As manufacturer, not independently audited, specifications, these are a transparent case study only. Assuming unity power factor, output is $100000x$ W; Equations 1 and 2 give Table 3.

Load / % rated	Output / kW	Core loss / W	Load loss / W	Total loss / W	Calculated efficiency / %
10	10.0	273	11	284	97.24
25	25.0	273	69	342	98.65
50	50.0	273	276	549	98.91
75	75.0	273	622	895	98.82
100	100.0	273	1105	1378	98.64

Table 3. Calculated from $P_{\text{loss}} = 273 + x^2(1105)$ and $\eta = P_{\text{out}} / (P_{\text{out}} + P_{\text{loss}}) \times 100$. Published loss values from [3]; remaining columns calculated by the author.

Total loss rises non-linearly from 284 W at 10% load to 1378 W at full load because the load component increases with x^2 . Efficiency instead peaks near 50%. Setting $273 = x^2(1105)$ gives $x = 0.497$, predicting 49.7% optimum load. The calculated 50% result, 98.91%, agrees with the DOE test condition for liquid-immersed units [1], supporting the model's physical basis.

The limitation is that this is a calculated case study based on one supplier's stated losses. It assumes unity power factor, rated sinusoidal voltage and constant temperature, while the actual losses of another 100 kVA transformer or a field-installed unit may differ. It is therefore sufficient to test the loss relationship, but not to estimate the whole population's operating efficiency. Its agreement with the independently specified 50% DOE test condition partly strengthens the model's credibility.

The dataset qualifies the claim. As $x \rightarrow 0$, the fixed core loss remains while useful output tends to zero, so efficiency tends to 0%, not 100%. Increasing load indefinitely also cannot improve efficiency because copper loss grows as $I^2 R$. Fractional load therefore supports the research question only when "ideal conditions" includes operation near the loss-balance point as well as very small loss mechanisms.

Review of sources and evidence quality

The strongest evidence is the DOE/eCFR material [1, 2]. It is a current primary regulatory source, not a peer-reviewed journal article, but it is authoritative for the performance standards being investigated. Its test method explicitly separates loss components, specifies reference temperatures and calibration, requires 0.01% efficiency resolution and limits power-loss measurement error to $\pm 3.0\%$ [1]. This makes the standard values highly suitable for controlled comparison. A compliance threshold is nevertheless not a random product sample or a real grid: annual efficiency also depends on load profile, temperature, power factor, harmonics, maintenance and age.

The textbook source [4] is older than the technical standards but is appropriate for stable electromagnetic theory rather than current performance data. Dataset 2 has high internal validity because class and load are constant, though it is prospective. The Larson Electronics specification [3] is the weakest source: it is commercial, undated and does not provide a laboratory procedure, uncertainty or independent certification. It is used only for its stated loss components, with all subsequent calculations and assumptions disclosed. The mutually consistent regulated data and loss model are credible enough to extrapolate cautiously to the claim about transformer conversion, but not to total grid efficiency.

Conclusion

For the research question, the relevant finding is that liquid-immersed transformer efficiency approaches 100% only when loss is a very small fraction of useful output. Capacity raised the regulated minimum from 98.70% at 10 kVA to 99.55% at 833 kVA. Tighter future standards raise the selected 833 kVA value to 99.64%, and the 100 kVA loss model predicts a 98.91% maximum near 49.7% load. These variables collectively support an approach to 100%, while every dataset retains a non-zero loss.

The sources have limitations, especially the single commercial loss specification and the standardised 50% test point, but the primary regulatory sources and established electromagnetic model are credible. In the ideal model, eliminating winding resistance, core hysteresis and eddy-current loss makes $P_{\text{loss}} = 0$ and $\eta = 100\%$. Based on the three investigated variables, capacity, design standard and fractional load, the claim is justified only as a limiting theoretical statement and as a close practical trend. It is misleading if it implies that a conventional operating transformer reaches exactly 100%.

Evaluation of the claim and improvements

The credible findings extrapolate to the original claim as follows: material and design improvements demonstrably move high-capacity transformers closer to 100%, but the persistent non-zero losses in all regulated standards prevent equality. This conclusion applies to electromagnetic transformer conversion only. It must not be extended to total grid efficiency, which additionally includes line losses, auxiliary equipment and multiple transformer stages.

The following improvements respond directly to the identified limitations:

1. Replacing the single supplier specification in Dataset 3 with independently certified test reports for at least five transformers across several ratings and manufacturers. This would enable uncertainty bars and distinguish manufacturer variation from the underlying load relationship.
2. Calculating an annual energy-weighted efficiency using a measured half-hourly load profile, rather than one 50% test point. This would test whether the nominal optimum matches actual operation.
3. Measuring no-load and load loss at several core and winding temperatures, with different harmonic distortion and power factors. This would quantify conditions that the controlled laboratory standard suppresses but a grid transformer experiences.

Extension

The scope can be extended by comparing amorphous-core and superconducting transformer designs with the same three variables: capacity, design standard and fractional load. A further extension would include total system efficiency by accounting for line loss, cooling or refrigeration power, and auxiliary equipment. Both extensions would broaden the evidence beyond conventional liquid-immersed units and give a more holistic evaluation of how closely an electricity-transmission system, rather than one transformer, can approach 100% efficiency.

References

1. U.S. Department of Energy, 2026, *10 CFR Part 431, Subpart K, Appendix A: Uniform Test Method for Measuring the Energy Consumption of Distribution Transformers*, Electronic Code of Federal Regulations, accessed 5 August 2026, <https://www.ecfr.gov/current/title-10/chapter-II/subchapter-D/part-431/subpart-K/appendix-Appendix%20A%20to%20Subpart%20K%20of%20Part%20431>.
2. U.S. Department of Energy, 2026, *10 CFR 431.196: Energy conservation standards and their effective dates*, Electronic Code of Federal Regulations, accessed 5 August 2026, <https://www.ecfr.gov/current/title-10/chapter-II/subchapter-D/part-431/subpart-K/section-431.196>.
3. Larson Electronics, n.d., *100 kVA overhead distribution transformer, pole mount: specification sheet*, accessed 5 August 2026, <https://www.larsonelectronics.com/images/product/Specsheet/244206.PDF>.
4. Sadiku, M.N.O., Alexander, C.K. & Musa, S. 2018, *Fundamentals of Electric Circuits*, 6th edn, McGraw-Hill Education, New York.